中英对照。英文为原文，中文为译文。类名、参数名、函数名、资源路径与网址保持英文，便于在虚幻引擎细节面板中定位。插图、标题和条目顺序与原 PDF 一致。




# FGear Overview

FGear 概述

FGear is a custom vehicle physics solution for unreal engine 4 that helps you build arcade/semi-arcade style racing games. The package consists of a core module and some extra features that were used for making the demo scenes. The package is mostly focused on 4 wheel cars but multi axle setups are also possible.
FGear 是面向虚幻引擎 4 的自定义载具物理方案，用来制作街机或半街机风格的竞速游戏。该包由一个核心模块，以及制作演示场景时使用的一些扩展功能组成。它主要针对四轮汽车，但也可以做多轴配置。

Core features:
核心功能：

- Desktop or mobile.
  桌面端或移动端。
- Adjustable integration steps.
  可调整的积分步长。
- Simplified pacejka, pacejka96 and MF6.1 tire model options.
  可选简化 Pacejka、Pacejka 96 和 MF6.1 轮胎模型。
- 3D wheel support(multiple raycasts or sphere/convex cast).
  三维车轮支持（多条射线检测，或球体/凸体投射）。
- Simple engine model with a torque curve.
  带扭矩曲线的简单发动机模型。
- Configurable auto/sequential/manual(h-pattern) transmissions.
  可配置的自动/序列式/手动（H 型换挡）变速箱。
- Open, locked or lsd differential options.
  开放式、锁止式或限滑差速器选项。
- Ackerman steering, toe and camber.
  阿克曼转向、前束和外倾。
- Adjustable torque and brake distribution of axles.
  可调整的车轴扭矩分配与制动分配。
- Simple spring/damper suspensions with preload option.
  带预载选项的简单弹簧/阻尼悬架。
- Multi axle setup support.
  支持多轴配置。
- AeroDynamics component for drag and downforce effects.
  用于阻力和下压力效果的 AeroDynamics 组件。

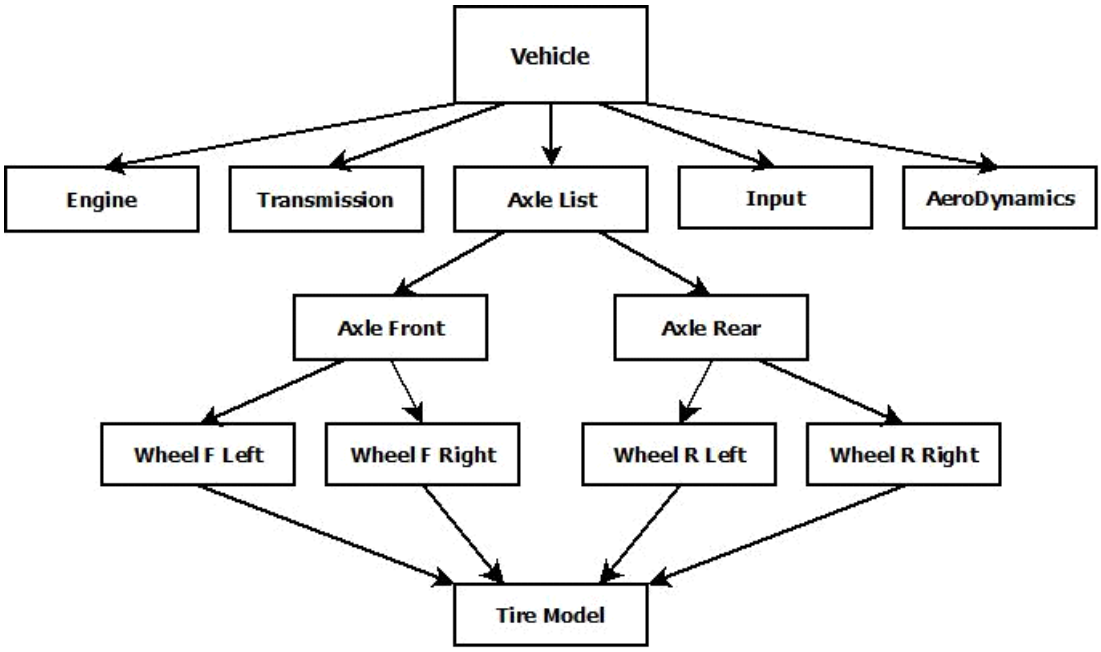
• Standard input manager with keyboard,joystick and wheel+shifter support.
标准输入管理器，支持键盘、手柄以及方向盘加换挡器。

• Driving aids like abs, asr, esp and anti roll bars.
驾驶辅助，例如 ABS、ASR、ESP 和防倾杆。

• Detailed telemetry ui.
详细的遥测界面。

• Various sample setups and demos.
多种示例配置和演示。

The below diagram shows how the core module is organized:
下图说明核心模块是如何组织的：




**• AFGearVehicle : This class is the host to all components of a vehicle and the only class that you need to inherit from for a driveable vehicle. Apart from managing components, this also contains implementations of most of the driving assists.**
AFGearVehicle：该类是一辆载具全部组件的宿主，也是做出可驾驶载具时唯一需要继承的类。除了管理组件，其中还包含大部分驾驶辅助的实现。

**• UFGearEngine : Uses a torque curve and rpm values to generate power to the vehicle.**
UFGearEngine：使用扭矩曲线和转速值为载具产生动力。

**• UFGearTransmission : This is a transmission that can be manually controlled or uses configurable values to automatically change gears for you. H-pattern or sequential shifting is possible and there is also a simple clutch implementation which is automatically managed in sequential mode.**
UFGearTransmission：这个变速箱可以手动控制，也可以用可配置的数值自动换挡。可以做 H 型换挡或序列式换挡，另外还有一个简单的离合器实现，在序列式模式下会自动管理。

**• UFGearAeroDynamics : This component has two jobs. First a drag force is generated as the vehicle goes faster and secondly a down force is applied. These are calculated according to the parameters and a constant air density value.**
UFGearAeroDynamics：该组件做两件事。第一，载具越快就产生越大的阻力；第二，施加下压力。二者按参数和一个恒定空气密度计算。

**• UFGearStandardInput : The first duty of this component is to read unreals input axis values and apply them to the vehicle like throttle or brakes. Keyboard, joystick or wheel+shifter inputs can be used(wheel hardware requires the raw input plugin of unreal). Optionally this modifies the steering inputs according to some parameters to provide a better control.**

UFGearStandardInput：该组件的首要职责是读取虚幻的输入轴数值，并作用到载具上，例如油门或制动。可以使用键盘、手柄或方向盘加换挡器（方向盘硬件需要虚幻的原始输入插件）。可选地，它会按某些参数修改转向输入，以提供更好的控制。

**• UFGearAxle : The main function of an axle is the power transfer. Engine torque is tranferred to the axle and the axle transfers torque to the wheels via a differential. A vehicle has 2 or more axles and an axle has 2 wheels attached.**
UFGearAxle：车轴的主要功能是传递动力。发动机扭矩传到车轴，车轴再通过差速器把扭矩传到车轮。一辆载具有 2 个或更多车轴，一个车轴连接 2 个车轮。

**• UFGearWheel : The wheel component both generates the tire forces and the suspension force. One or more raycasts are used to determine the suspension compression and the required force is calculated to keep the vehicle away from ground. To generate lateral and longitudinal forces, a tire model is used. The generated forces are applied to the vehicle body and also the reaction torques are fed back to the engine.**
UFGearWheel：车轮组件既产生轮胎力，也产生悬架力。用一条或多条射线检测确定悬架压缩量，并算出使载具离开地面所需的力。横向力和纵向力由轮胎模型产生。产生的力作用到车身上，反作用力矩也会反馈给发动机。

**• UFGearTire : Wheels use tire model(s) to calculate traction forces. The default tire model is pacejka96. Simplified pacejka is also decent and computationaly cheaper. MF6.1 tire model is the most advanced one. Some features like camber effect is not available with the simple model. Which one to use depends on your requirements. You can read more about tire models in CoreModule documentation.**
UFGearTire：车轮使用轮胎模型计算牵引力。默认轮胎模型是 Pacejka 96。简化 Pacejka 也不错，而且计算更省。MF6.1 是最完整的一种。外倾效果等部分功能在简单模型中不可用。用哪一种取决于需求。轮胎模型的更多内容见 CoreModule 文档。

**• UFGearReplication : Aims to provide smooth network replication. It has three main tasks, input and state synchronization and client prediction.**
UFGearReplication：旨在提供平滑的网络复制。它有三项主要任务：输入同步、状态同步和客户端预测。

**• Extra Features(Optional) : There are some additional content that are used for the sample scenes. You can use them for prototyping and learning purposes but they are not recommended for production use. Here are the extras:**
Extra Features（可选）：有一些用于示例场景的附加内容。可以用来做原型和学习，但不建议用于正式产品。扩展内容如下：

**• UFGearOrbitCamera : A simple configurable camera component that orbits around the vehicle.**
UFGearOrbitCamera：一个可配置的简单摄像机组件，绕载具环绕。

**• UFGearEffects: First of all this component adds engine and skidding sound effects along with a AFGearSkidMark mesh effect and smoke particles. Additionally this contains a sample system that detects each wheels hit-surface physical material(you need to use FGearPhysicalMaterial) and alters friction values according to provided parameters.**
UFGearEffects：该组件首先添加发动机声和打滑声，以及 AFGearSkidMark 网格效果和烟雾粒子。另外它还包含一个示例系统，检测每个车轮命中表面的物理材质（需要使用 FGearPhysicalMaterial），并按给定参数改变摩擦值。

**• UFGearArcadeAssists : This component contains a couple of fake but effective driving assists. These can be useful for arcade style games.**
UFGearArcadeAssists：该组件包含若干不真实但有效的驾驶辅助。它们可用于街机风格的游戏。

**• UFGearCustomCollider : This enables you to separate collision from inertial properties of the vehicles physics body.**
UFGearCustomCollider：它让你把碰撞与载具物理刚体的惯性属性分开。

**• AFGearRewindReplay : A simple actor that can be placed in a map to track vehicles history and rewind or**

**replay when needed.**
AFGearRewindReplay：一个可以放到地图中的简单 Actor，用来记录载具历史，并在需要时倒带或回放。

For a quick start go to 1-Quickstart or check our youtube channel https://www.youtube.com/playlist?list=PL0io2GoXo8vx_3Ep4dNMLcvD0OyWwUGMm
快速开始请看 1-Quickstart，或查看我们的 YouTube 频道 https://www.youtube.com/playlist?list=PL0io2GoXo8vx_3Ep4dNMLcvD0OyWwUGMm

For a detailed explanation of core module go to 2-CoreModule.
核心模块的详细说明见 2-CoreModule。

For a detailed explanation of extras go to 3-Extras.
扩展功能的详细说明见 3-Extras。

For any kind of support contacts us at lazybitgames@gmail.com
如需任何支持，请联系 lazybitgames@gmail.com

# Changelog
更新日志

v1.8.5
v1.8.5

- Unreal 5.8 support
  支持 Unreal 5.8
- Improvements and bug fixes
  改进与错误修复

Notes
说明

- Dropped 5.5 support.
  不再支持 5.5。
- Cascade particles are replaced with niagara particles.
  Cascade 粒子被替换为 Niagara 粒子。
- Finar gear ratio parameter will be set per axle now. This solves the stability issues when different axles have different wheel radius.
  Finar gear ratio（最终传动比）参数现在按车轴分别设置。这样可以解决不同车轴车轮半径不同时的稳定性问题。
- Suspension force is scaled with hit normal dot to prevent high angle issues (disabled for bikes).
  悬架力会乘以命中法线的点积，以防止大角度问题（对摩托车禁用）。
- Engine brake definition changed from zero throttle to negative torque.
  发动机制动的定义从零油门改为负扭矩。
- Refactored wheel rolling friction code.
  重构了车轮滚动摩擦代码。
- Vehicle.getMinWheelKMHSpeed function is fixed to return a signed speed which solves a rare gear change issue (going backward at 1st gear then pressing back will brake instead of changing to R).
  修正了 Vehicle.getMinWheelKMHSpeed，使它返回带符号的速度。这解决了一个少见的换挡问题：在一挡向后移动时再按后退，会制动而不是换到 R。
- Updated default drivetrain friction value.
  更新了默认的传动系统摩擦值。

• Removed suspension constraint component.
移除了悬架约束组件。

v1.8.4
v1.8.4

• Unreal 5.7 support
支持 Unreal 5.7

• Ability to change vehicle mesh at runtime
可以在运行时更换载具网格

• Improvements and bug fixes
改进与错误修复

Notes
说明

• Fixed the problem where the ordering of axles was shuffled. This could cause issues when you have 4 or more axles. The axles are sorted based on their distance to the front axle now. Additionally you can call Vehicle.setAxleIndex to alter the indices at runtime.
修正了车轴顺序被打乱的问题。有 4 个或更多车轴时这会造成问题。现在车轴按其到前轴的距离排序。另外可以在运行时调用 Vehicle.setAxleIndex 修改索引。

• Removed the call to mesh->SetAllBodiesSimulatePhysics(true) in vehicle beginplay. In certain cases you may not want a vehicle part to be simulated so from now on users should manage all the bodies themselves.
移除了载具 BeginPlay 中对 mesh->SetAllBodiesSimulatePhysics(true) 的调用。某些情况下你可能不希望某个载具部件被模拟，因此从现在起应由用户自己管理全部刚体。

• Fixed the issue where time dilation could break the simulation.
修正了时间膨胀可能破坏模拟的问题。

• Fixed the order of torque components in MF6.1 tire model.
修正了 MF6.1 轮胎模型中扭矩分量的顺序。

• Fixed FGearUtility::Euler function which caused some rotations like camber to be inverted.
修正了 FGearUtility::Euler，该函数曾使外倾等某些旋转反向。

• Vehicle's Super::Beginplay call is moved to the end of the beginplay function. This may prevent some crash cases.
载具的 Super::BeginPlay 调用被移到 BeginPlay 函数末尾。这可以避免一些崩溃。

• Reference camber value is added to tire models. You can preview the effect of camber angle in the tire model blueprints now.
轮胎模型增加了参考外倾值。现在可以在轮胎模型蓝图中预览外倾角的效果。

• Updated tire96 and mf6.1 tire models. Certain inputs and intermediate values are clamped based on original equations.
更新了 Tire96 和 MF6.1 轮胎模型。某些输入和中间值会按原始公式进行钳制。

• MaxHillHolderSpeed variable in now serialized, can be saved/load via json.
MaxHillHolderSpeed 变量现在会被序列化，可以通过 JSON 保存和加载。

• Fixed the bug where partial throttle + partial brake could lead to faster acceleration when ABS is on.
修正了 ABS 开启时，部分油门加部分制动可能导致加速更快的错误。

• Effect components tick group is set to post physics now.
效果组件的 Tick 组现在设为 Post Physics。

• Improved torque.Y calculation in wheel class. Instead of blindly checking for brake input we now make sure the vehicle is decelerating.

改进了车轮类中 torque.Y 的计算。不再盲目检查制动输入，而是确认载具正在减速。

• Improved activation/deactivation of hill holder. It won't activate on a flat surface and it can deactivate if the slope angle changes after activation.
改进了坡道保持的启用和停用。它不会在平地上启用，并且启用后若坡度角变化也可以停用。

• Vehicle.setMesh function can be used to change the vehicle mesh. You need to provide the new mesh, animation bp and physics asset references. If the bone names differ then you also need to change the bone name parameter of each wheel.
可以用 Vehicle.setMesh 更换载具网格。需要提供新的网格、动画蓝图和物理资产引用。若骨骼名不同，还需要修改每个车轮的骨骼名参数。

• Fixed a problem with the auto clutch where it failed for the rear gear in manual transmission mode.
修正了自动离合器在手动变速箱模式下对倒挡失效的问题。

v1.8.3
v1.8.3

• Unreal 5.6 support
支持 Unreal 5.6

• Hill-holder assist
坡道保持辅助

• Vertical offset option for wheels
车轮垂直偏移选项

• Improvements and bug fixes
改进与错误修复

Notes
说明

• Hillholder does not work like in real cars, you do not need to brake in order to initiate the hold. When you come to a stop on a hill the vehicle will brake on it's own before it starts rolling back. The side effect is that it also engages when you come to a stop on a flat surface but it's harmless.
坡道保持不像真实汽车那样工作，不需要先制动才能进入保持。在坡上停住后，载具会在开始后溜之前自己制动。副作用是在平地上停住时也会介入，但没有害处。

• Renamed UFGearWheel. setLateralOffsets function to UFGearWheel. setWheelOffsets. You can use it to update all wheel offsets at runtime.
把 UFGearWheel.setLateralOffsets 重命名为 UFGearWheel.setWheelOffsets。可以用它在运行时更新全部车轮偏移。

• Fixed the issue where an unpossessed vehicle would be teleported to zero origin. While this behavior is fixed developers should make sure a vehicle is never unposssed in multiplayer. A pawn without a controller can not send it's data to the server.
修正了未被 Possess 的载具会被传送到零原点的问题。该行为已修复，但开发者仍应确保多人游戏中载具从不被取消 Possess。没有控制器的 Pawn 无法把数据发给服务器。

• A new event is fired from the engine component called OnStartStop which explains itself.
发动机组件新增一个名为 OnStartStop 的事件，其含义从其名称即可看出。

• A new option called "Query Response Overrides" is added to the tracing options. With this you can create exceptions for the query responses of line and sphere queries(not supported with convex casting). This is equivalent of the FCollisionResponseParams option in C++.
追踪选项中新增 "Query Response Overrides"。可以用它为线段和球体查询的响应创建例外（凸体投射不支持）。这相当于 C++ 中的 FCollisionResponseParams 选项。

• An empty physics asset won't crash the engine anymore.

空的物理资产不再会使引擎崩溃。

• Removed “Allow Wheel Rotation Against Engine” option.
移除了 “Allow Wheel Rotation Against Engine” 选项。

• Dropped 5.3 support.
不再支持 5.3。

v1.8.2
v1.8.2

• Unreal 5.5 support
支持 Unreal 5.5

• Engine stalling (experimental)
发动机熄火（试验性）

• Improvements and bug fixes
改进与错误修复

Notes
说明

• Engine stalling is only meant for manual transmission/clutch mode. You need to set a stall rpm and stall time, when the engine is forced to go below that rpm for enough time then the engine is forced to stop. A negative stall time value will prevent stalling.
发动机熄火只用于手动变速箱/离合器模式。需要设置熄火转速和熄火时间；当发动机被迫低于该转速并持续足够时间后，发动机会被强制停止。熄火时间为负时不会熄火。

• Removed a problematic clutch axis from the example project which kept the clutch pressed with a controller.
从示例工程中移除了一个有问题的离合器轴，它会在使用控制器时一直踩住离合器。

• Fixed the case where engine doesn’t rev-up when started in neutral gear.
修正了在空挡启动时发动机不拉高转速的情况。

• Pause button added to replay sample so it’s possible to pause the replay now. Also it is possible to access the variables of this class from blueprints.
回放示例增加了暂停按钮，因此现在可以暂停回放。也可以从蓝图访问该类的变量。

• Forced symmetry on all sample tire models eliminating left/right steering balance issues.
对所有示例轮胎模型强制对称，消除了左/右转向不平衡。

• Fixed warnings in the serialization class.
修正了序列化类中的警告。

• Dropped 5.2 support.
不再支持 5.2。

v1.8.1
v1.8.1

• Enhanced input bugs fixed
修正了增强输入的错误

• Some replication bugs fixed (read notes)
修正了部分复制错误（见说明）

• Some minor improvements (read notes)
一些小改进（见说明）

Notes
说明

• When we had more then 4 enhanced input actions bindings it was possible to loose references to some of these bindings. We changed the way we store action bindings and the issue is gone.
当增强输入动作绑定超过 4 个时，可能会丢失其中一些绑定的引用。我们改变了保存动作绑定的方式，该问题已消失。

• We used to only have 3 sample input actions for enhanced input in the sample project. Now we added input actions for all input types of the standard input. These files are under FGearAssets/Inputs folder in the sample project.
示例工程中原先只有 3 个增强输入的示例输入动作。现在为标准输入的全部输入类型都添加了输入动作。这些文件位于示例工程的 FGearAssets/Inputs 文件夹。

• Compression of the gear input was failing with negative numbers in replication. This was an issue when you have a sequential gearbox, this is fixed now.
复制时，挡位输入的压缩在负数上会失败。使用序列式变速箱时会有这个问题，现已修正。

• Instead of replicating the gear input, we now replicate the current gear value. This fixes the gear sync issues with seuqential and manual transmissions.
现在复制的是当前挡位值，而不是挡位输入。这修正了序列式和手动变速箱的挡位同步问题。

• Current input struct is no longer replicated back to it's owner. Since the input struct is updated every tick this was not an issue but we skipped an unnecessary transfer.
当前输入结构不再复制回它的拥有者。因为输入结构每 Tick 都会更新，这本来不是问题，但我们省去了一次不必要的传输。

• There was an issue with server authoritative replication in 5.4. It seems moving a kinematic object no longer generates velocity by default. We had to activate this behavior in 5.4.
5.4 中服务器权威复制有一个问题。移动运动学对象似乎默认不再产生速度。我们不得不在 5.4 中启用这一行为。

• Clutch input was skipped when you have auto-clutch. From now on if you press the clutch it will override any auto-clutch value. This way you can let the system run the clutch and override whenever you want.
启用自动离合器时会跳过离合器输入。从现在起，如果踩下离合器，就会覆盖任何自动离合器的值。这样可以让系统运行离合器，并在需要时覆盖它。

• Wheels can generate inherent and extra(additional) drag forces now. The idea is to let a wheel drag in certain cases like a flat tire or when a tire is on rough terrain like mud or sand. The inherent drag is supposed to be the internal drag of a wheel and it is zero by default. The additional drag is supposed to be updated at runtime depending on some conditions. These two parameters will be summed up and used as the coefficient of the wheels drag force. As an example the effects component checks the physical material of the surface that each wheel drives on and sets the additional drag based on the roughness parameter of the physics material. In the sample project you will observe that the vehicle struggles more on the terrain. You can use Wheel.setInherentDrag & Wheel.setAdditionalDrag functions to modify these parameters. This is an experimental feature, use with caution.
车轮现在可以产生固有阻力和额外（附加）阻力。目的是让车轮在某些情况下产生阻力，例如爆胎，或轮胎处在泥地、沙地这类粗糙地形上。固有阻力应当是车轮的内部阻力，默认是零。附加阻力应当在运行时按某些条件更新。这两个参数会相加，作为车轮阻力的系数。例如效果组件会检查每个车轮所行驶表面的物理材质，并按物理材质的粗糙度参数设置附加阻力。在示例工程中可以看到载具在地形上更吃力。可以用 Wheel.setInherentDrag 和 Wheel.setAdditionalDrag 修改这些参数。这是试验功能，请谨慎使用。

v1.8
v1.8

• Unreal5.4 support
支持 Unreal 5.4

• Client authoritative replication

客户端权威复制

• Enhanced input support
支持增强输入

• Rev-matching for manual transmission
手动变速箱的转速匹配

• Replicated sleeping
可复制的休眠

• Forward offset option for wheels
车轮前向偏移选项

Notes
说明

• Checkout CoreModule.pdf document for a more detailed explanation of the new replication options.
新复制选项的更详细说明请查看 CoreModule.pdf。

• In client authoritative mode all the vehicles are simulating physics and the control over your vehicle is just like a standalone game. While the input replication is quite similar to the server authoritative mode, state replication is quite different. It is a combination of a network correction and temporary physics blending after collisions. We tried to handle collisions gracefully but you can still encounter bad collision reactions with bad network conditions.
在客户端权威模式下，所有载具都在模拟物理，对自己载具的控制就像单机游戏。输入复制与服务器权威模式相当相似，但状态复制差别很大。它是网络校正与碰撞后临时物理混合的组合。我们尽量妥善处理碰撞，但网络状况很差时仍可能遇到糟糕的碰撞反应。

• Client authoritative replication can blend replication transform with the physics transform. For this to happen we listen to the NotifyHit callbacks in the vehicle class and trigger the blend if the conditions are met. If the collision profiles of the vehicles are not properly setup then this blend may not be triggered at all and it's difficult tell whether if it's working or not. In order to diagnose such issues we introduced a new telemetry option called "Print Collision Logs". When this is on you should see a log printed whenever two vehicles collide at a speed greater then MinCollisionBlendingSpeed. If you do not see the logs you can tell that there is a problem with the collision(hit event) setups. This log is only printed with the client authoritative replication mode.
客户端权威复制可以把复制变换与物理变换混合。为此我们在载具类中监听 NotifyHit 回调，并在条件满足时触发混合。若载具的碰撞预设没有正确设置，该混合可能完全不会触发，而且很难判断它是否在工作。为了诊断这类问题，我们增加了名为 "Print Collision Logs" 的遥测选项。启用后，两辆载具以大于 MinCollisionBlendingSpeed 的速度碰撞时，应当看到一条日志。如果看不到日志，就可以判断碰撞（命中事件）设置有问题。该日志只在客户端权威复制模式下打印。

• Existing server authoritative replication is still available but it is off by default. It can be turned back on via setttings in ProjectSettings > Game > FGearCommon.
原有的服务器权威复制仍然可用，但默认关闭。可以通过 Project Settings > Game > FGearCommon 中的设置重新打开。

• Each vehicle had its own replication settings and we had the option to override and use common settings(ProjectSettings > Game > FGearCommon). From now on we use the common settings by default and each vehicle can override the settings if needed.
每辆载具原先都有自己的复制设置，并且可以选择覆盖并使用共用设置（Project Settings > Game > FGearCommon）。从现在起默认使用共用设置，每辆载具可以在需要时覆盖这些设置。

• Removed buffered input replication. This prevents any input delay due to server slowdowns.
移除了带缓冲的输入复制。这可以避免服务器变慢造成的输入延迟。

• Reduced default input send rate from 50 to 20 same as state send rate.

默认输入发送频率从 50 降到 20，与状态发送频率相同。

• Input and state replication variables are now compressed/quantized for reducing bandwidth costs.
输入和状态复制变量现在会压缩/量化，以降低带宽开销。

• Input and state replications are longer optional but you can set zero send rate for each to pause the RPCs.
输入复制和状态复制不再是可选项，但可以把各自的发送频率设为 0 来暂停 RPC。

• Added an AI driver following a spline loop in the Multiplayer map.
多人地图中增加了一个沿样条闭环行驶的 AI 驾驶员。

• Auto drive is no longer a component but an AIController. Instead of adding as a component you need to possess the vehicle with this controller. The controller won't function with its default form so it's better to derive from it and create a BP(there is a sample bp called FGearAutoDrive_BP in the sample project). Spawn the controller(you can both manually spawn or let the engine spawn it), tune it's settings, provide a spline to follow(FGearAutoDrive_BP looks for the first spline in the map and assigns it at beginplay) and finally possess the vehicle. This is exemplified in the AITest map.
自动驾驶不再是组件，而是一个 AIController。不要再把它作为组件添加，而要用该控制器 Possess 载具。控制器的默认形态不会工作，因此最好从它派生并做一个蓝图（示例工程中有名为 FGearAutoDrive_BP 的示例蓝图）。生成该控制器（可以手动生成，也可以让引擎生成），调整它的设置，提供一条要跟随的样条（FGearAutoDrive_BP 会在 BeginPlay 时查找地图中的第一条样条并指定给自己），最后 Possess 载具。AITest 地图中有示例。

• Auto drive inputs were applied directly to the vehicle components before but now the inputs are applied via the standard input component. This made it possible to use it with both server & client authoritative replication modes.
自动驾驶的输入以前直接作用到载具组件上，现在通过标准输入组件施加。这使得它既能用于服务器权威，也能用于客户端权威复制模式。

• Auto drive is no longer ticking by its own but it is updated by the onOverrideInput delagate which is explained below. This let's the AI inputs to be replicated properly.
自动驾驶不再自己 Tick，而是由下面说明的 onOverrideInput 委托更新。这样 AI 输入才能被正确复制。

• Removed overrideInputTick native event and added onOverrideInput delegate instead. This delegate is either called in the actor tick or the replication tick so we can also override inputs in multiplayer.
移除了 overrideInputTick 原生事件，改为 onOverrideInput 委托。该委托在 Actor Tick 或复制 Tick 中调用，因此在多人游戏中也可以覆盖输入。

• Mobile input sample now uses the onOverrideInput delegate to inject UI inputs.
移动端输入示例现在使用 onOverrideInput 委托注入界面输入。

• Replication component used to override input-reading option of standard input. This is no longer needed and is not mandatory for input overriding.
复制组件以前会覆盖标准输入的读取输入选项。现在不再需要，输入覆盖也不再强制依赖它。

• Checkout CoreModule.pdf document(StandardInput settings) for a more detailed explanation of how you can activate and use enhanced input system.
如何启用和使用增强输入系统的更详细说明，请查看 CoreModule.pdf（StandardInput 设置）。

• We have added some basic enhanced input assets under FGearAssets/Inputs folder in the sample project.
我们在示例工程的 FGearAssets/Inputs 文件夹中添加了一些基础的增强输入资产。

• Raw input values can be accessed via BP now. This lets you keep current player inputs and modify/alter some of them.
现在可以通过蓝图访问原始输入值。这样可以保留当前玩家输入，并修改其中一部分。

• Open differential implementation used to check the wheel rotation speeds but it is only based on wheel load right now which is better at imitating an open diff.
开放式差速器的实现以前检查车轮转速，现在只基于车轮载荷，这更能模仿开放式差速器。

• Torque bias ratio was unlimited for locking diff calculation, it is limited to 9:1 now.
锁止差速计算的扭矩偏置比以前没有上限，现在限制为 9:1。

• Manual transmission rev-matching is only for downshifting.
手动变速箱的转速匹配只用于降挡。

• A sleeping vehicle couldn’t wake up in a multiplayer game, this is fixed now.
休眠中的载具在多人游戏里无法唤醒，现已修正。

• Users can now query the visual location and rotation of each wheels with UFGearWheel.getWheelPosition and UFGearWheel. getWheelRotation functions.
用户现在可以用 UFGearWheel.getWheelPosition 和 UFGearWheel.getWheelRotation 查询每个车轮的视觉位置和旋转。

• Forward offset of a wheel can be updated at runtime with UFGearWheel. setLateralOffsets function.
可以用 UFGearWheel.setLateralOffsets 在运行时更新车轮的前向偏移。

• Added a separate option to mute the engine sound when the vehicles goes to sleep.
增加了一个单独选项，在载具进入休眠时静音发动机声。

• Added a config line(AspectRatioAxisConstraint) to the DefaultEngine.ini which prevents the camera fov change that started happening after 5.3.
在 DefaultEngine.ini 中增加了一行配置（AspectRatioAxisConstraint），用来防止 5.3 之后开始出现的摄像机视野变化。

• Fixed a crash case which happens when you put a vehicle in a sublevel and stream out in unreal5.
修正了把载具放进子关卡并在 Unreal 5 中流送卸载时发生的崩溃。

v1.7
v1.7

• Reworked powertrain
重做了动力传动系统

• Sleeping (auto/manual)
休眠（自动/手动）

• Rev-matching option for transmission
变速箱的转速匹配选项

• Stall assist option for clutch
离合器的熄火辅助选项

• Spacer offset option for wheels
车轮的轮距垫片偏移选项

• Simple cruise control with a PID controller
使用 PID 控制器的简单巡航控制

Notes
说明

• From now on the changelog can be found in the “0-Overview.pdf” document
从现在起，更新日志位于 “0-Overview.pdf” 文档中

• All gettters are now const functions, you might need to reiterate your blueprints
所有 getter 现在都是 const 函数，可能需要重新整理蓝图

• All the sample vehicles have more realistic wheel mass values now
所有示例载具现在都使用更真实的车轮质量

• All the sample vehicles’ linear and angular damping values are updated. Even having a small linear damping

value can change the acceleration performance of the vehicle especially at high speeds
更新了所有示例载具的线阻尼和角阻尼。即使很小的线阻尼也会改变载具的加速表现，高速时尤其明显

• Clutch value is also displayed on the gauge telemetry now
仪表遥测现在也会显示离合器值

• An option called optimum slip weight is added, this helps to reduce the speed mismatch between the engine and wheel rpms. This is an experimental feature and can affect the overall driving behavior so be cautious with it.
增加了名为 optimum slip weight 的选项，用来减小发动机转速与车轮转速的不匹配。这是试验功能，可能影响整体驾驶表现，请谨慎使用。

• Fixed the case where 4wd vehicles failed to generate correct torque output
修正了四驱载具无法产生正确扭矩输出的情况

• You can manually put a vehicle into sleep and/or wake it by calling sleep/wakeUp functions. This is demonstrated in the dragrace map, all the vehicles are put to sleep at the finish line.
可以调用 sleep/wakeUp 手动让载具休眠和/或唤醒。DragRace 地图中有演示，所有载具在终点线处被设为休眠。

• Added a new sample scene that demonstrates the auto sleeping of vehicles. Sleeping vehicles color turn to black and they go back to original when they wake up. If you set a minumum linear and angular speed threshold then the vehicle will automatically sleep when the conditions are met. You can also define the set of inputs that will wake the vehicle.
增加了一个演示载具自动休眠的新示例场景。休眠载具的颜色变为黑色，唤醒后恢复原色。如果设置最低线速度和角速度阈值，条件满足时载具会自动休眠。也可以定义能唤醒载具的一组输入。

• A vehicle that is touching another simulated actors bounds can’t go to sleep. This fixes a set of problems where we could just penetrate into other vehicles when they are asleep.
正在接触另一个被模拟 Actor 包围体的载具不能进入休眠。这修正了一组问题：以前载具休眠时，我们可能直接穿进其他载具。

• A sleeping vehicle will pause its sfx and particle effects. Muting the engine sound is optional.
休眠中的载具会暂停其音效和粒子效果。是否静音发动机声是可选的。

• Cruise control and the PID controller options are located in the ArcadeAssists component
巡航控制和 PID 控制器选项位于 ArcadeAssists 组件中

• There is a cruise speed option in the modification map gui, you can set a target speed and the vehicle will try to maintain it
修改地图的界面中有巡航速度选项，可以设置目标速度，载具会尝试维持它

• Fixed the ArcadeAssists::setCruiseSpeed function, it was just not working correctly
修正了 ArcadeAssists::setCruiseSpeed，它之前不能正确工作

• Fixed the case where the MobileInput map could fail to get the reference of a different vehicle other then the vehicle on the map
修正了 MobileInput 地图可能无法取得地图上那辆车以外的其他载具引用的情况

• Fixed a bug with the tire curve previews
修正了轮胎曲线预览的一个错误

• Fixed a bug where the top speed of each gear ratio was miscalculated at editor time
修正了编辑器中每个齿比最高速度计算错误的问题

• Brake & clutch engagement curves are moved to standard input and also added a throttle engagement curve
制动和离合器接合曲线移到了标准输入，并增加了油门接合曲线

• Gear up and gear down velocity bias options are added, this way it will be possible to change gears based on wheel speed rather then velocity

增加了升挡和降挡的速度偏向选项，这样就可以按车轮转速而不是按速度换挡

• Added drivetrain inertia option which is used in the wheel rotation calculations. This has a similar effect to the wheel mass option, larger inertia means slower accelerations
增加了传动系统惯量选项，用于车轮转动计算。效果与车轮质量选项相似，惯量越大加速越慢

• Added drivetrain friction option which is a new factor that slows down the vehicle. Unlike wheel roll resistance this is a constant factor
增加了传动系统摩擦选项，这是使载具变慢的一个新因素。与车轮滚动阻力不同，它是一个常数因子

• Reverse brake option added, if you press throttle while going backwards in 1st gear the vehicle will automatically brake for you. The same thing works with the reverse gear.
增加了反向制动选项。在一挡向后行驶时踩油门，载具会自动制动。倒挡也是如此。

• Spacer offset will add distance between the left and right wheels, resulting a wider track. Do not confuse it with the rim offset which only moves the physics representation.
轮距垫片偏移会增加左右车轮之间的距离，从而加宽轮距。不要把它和轮辋偏移混淆，轮辋偏移只移动物理表现。

• A new set of vehicles are included in the example project now, these are from Epic's free content and they have slightly different setups.
示例工程现在包含一组新载具，它们来自 Epic 的免费内容，配置略有不同。

• A new sandbox map is added to the example project with the new vehicles including a truck jointed to its trailer.
示例工程增加了一张新的沙盒地图，其中有这些新载具，包括一辆与拖车铰接的卡车。

• Added minumum ABS speed parameter, below this speed ABS will not work. This is added to avoid a problem where the brakes fail to fully stop the vehicle with ABS.
增加了 ABS 最低速度参数，低于该速度时 ABS 不工作。这是为了避免 ABS 开启时制动无法让载具完全停住的问题。

v1.6.2
v1.6.2

• Unreal5.2 support
支持 Unreal 5.2

• Improvements and bug fixes
改进与错误修复

Notes
说明

• Ensure fails are eliminated for non-substep mode.
非子步模式下不再出现 Ensure 失败。

• Overall refactor to use physics transform instead of actor transform for vehicle calculations. This fixes some hidden issues and makes the plugin more stable in Unreal5.
整体重构为在载具计算中使用物理变换，而不是 Actor 变换。这修正了一些隐蔽问题，并使插件在 Unreal 5 中更稳定。

• A landscape is added to sandbox map.
沙盒地图增加了地形。

• Added an option to override CollisionTrace for custom collision in order to fix Unreal5 issues.
为自定义碰撞增加了覆盖 CollisionTrace 的选项，以修正 Unreal 5 的问题。

• Fixed a minor issue with the Constraint map.
修正了 Constraint 地图的一个小问题。

• Engine power calculation wasn't taking friction torque into account, this is fixed now.
发动机功率计算以前没有计入摩擦扭矩，现已修正。

• Added ability to set bone name of wheels.
增加了设置车轮骨骼名的能力。

• Convex casting meshes no longer generate overlap events with the world.
凸体投射网格不再与世界产生重叠事件。

• Wheel.setRadius function is updated to work with convex casting.
更新了 Wheel.setRadius，使其适用于凸体投射。

• Convex casting options are extended and moved to vehicle class.
凸体投射选项已扩展，并移到载具类中。

• A custom cylinder mesh asset is included in the plugin folder which works fine against the landscape in Unreal5.
插件文件夹中包含一个自定义圆柱网格资产，它在 Unreal 5 中对地形表现正常。

• Cleared all deprecated warnings.
清除了全部弃用警告。

v1.6.1
v1.6.1

• Unreal5.1 support
支持 Unreal 5.1

• Minor bug fixes
少量错误修复

Notes
说明

• Dropped 4.26 support
不再支持 4.26

• Fixed crash case when async callback is not properly unregistered during exit
修正了退出时异步回调没有正确注销而导致的崩溃

• Fixed differential bug (lsd/locked options were reversed)
修正了差速器错误（限滑/锁止选项被对调）

• Fixed the issue with custom UI
修正了自定义界面的问题

• SetWorldScale3D replaced with SetRelativeScale3D for convex casting mesh
凸体投射网格用 SetRelativeScale3D 替换了 SetWorldScale3D

• Convex casting against landscape is still buggy
对地形做凸体投射仍然有错误

v1.6
v1.6

• Unreal5 support
支持 Unreal 5

• Custom gravity option
自定义重力选项

• Custom gravity sample map

自定义重力示例地图

• Better hard contact implementation
更好的硬接触实现

• Ability to offset casting origin
可以偏移投射起点

• Improvements and bug fixes
改进与错误修复

Notes
说明

• Due to unsolved/undiscovered issues with chaos the overall driving behavior might be different then UE4.
由于 Chaos 中尚未解决或尚未发现的问题，整体驾驶表现可能与 UE4 不同。

• Without substepping you can get ensure fails in the log with chaos.
不使用物理子步时，Chaos 下日志中可能出现 Ensure 失败。

• Use custom gravity with chaos or your vehicles can freeze/stop in air for no reason.
在 Chaos 下请使用自定义重力，否则载具可能无故在空中冻结或停住。

• All sample vehicles are set to use custom gravity.
所有示例载具都设为使用自定义重力。

• Convex casting used to generate lateral suspension forces, they are eliminated for correct behavior.
凸体投射以前会产生侧向悬架力，为了得到正确表现，这些力已被去掉。

• Blocking hits are forced for all raycasts/sweeps.
所有射线检测和扫描都强制为阻挡命中。

• The sphere used in custom gravity map does not match its collider so wheels can penetrate the sphere.
自定义重力地图中使用的球体与其碰撞体不匹配，因此车轮可能穿透该球体。

• Added an option to override collision channel for custom colliders.
增加了为自定义碰撞体覆盖碰撞通道的选项。

• Somehow default WorldDynamic channel does not collide with the world with chaos so we set it to WorldStatic in the samples.
默认的 WorldDynamic 通道在 Chaos 下不知为何不与世界碰撞，因此示例中把它设为 WorldStatic。

• New hard contact scale parameter can behave jumpy with high values, use a reasonable value like 0.2-0.3.
新的硬接触缩放参数在数值很高时可能跳动，请使用 0.2 到 0.3 这类合理数值。

• The order of brake/engine torque generation is swapped. If engine torque is more then brakes then the wheels will spin. This will make it easier to make burnouts.
制动扭矩和发动机扭矩的生成顺序已对调。若发动机扭矩大于制动，车轮就会空转。这会让烧胎更容易。

• All force/torque/impulses are accumulated and applied at the same time now.
所有力、扭矩和冲量现在会先累加，再同时施加。

• If your wheels get stuck under the ground when the vehicle falls from height then you can use the cast offset parameter. A positive value will offset the origin of the casts along the suspension normal. Use a reasonable number to keep the origins inside the vehicle hull. You can activate the telemetry gizmo to view the origins.
若载具从高处落下后车轮卡在地面以下，可以使用投射偏移参数。正值会沿悬架法线偏移投射起点。请使用合理数值，使起点留在车身外壳内部。可以打开遥测 Gizmo 查看这些起点。

• High friction engines could generate negative torques and the vehicle could travel backwards, this is fixed now.

高摩擦的发动机可能产生负扭矩，载具可能向后行驶，现已修正。

• 4.25 support is dropped, epic do not allow more then 3 versions to be uploaded.
不再支持 4.25，Epic 不允许上传超过 3 个版本。

v1.5
v1.5

• Convex casting option
凸体投射选项

• Wheel reaction forces
车轮反作用力

• Ability to turn engine on/off
可以打开或关闭发动机

• Improvements and bug fixes
改进与错误修复

Notes
说明

• By default a cylinder mesh from "EngineContent/BasicShapes" is used for convex casting. A query only hidden mesh object is created for this purpose. If you want to provide another mesh make sure its scale and orientation matches that cylinder. Convex casting is more sensitive to floating point precision issues so for ex. if you use a large(a couple of km wide) plane/cube/brush as ground then casts can return unexpected results and vehicles can shake wildly for no reason.
默认使用 “EngineContent/BasicShapes” 中的圆柱网格做凸体投射。为此会创建一个仅用于查询的隐藏网格对象。若要提供另一个网格，请确保其缩放和朝向与该圆柱一致。凸体投射对浮点精度问题更敏感。例如用一个很大（宽约数公里）的平面、立方体或笔刷作为地面时，投射可能返回意外结果，载具会无故剧烈抖动。

• Wheel reaction force is enabled with a float instead of a boolean. Wheel forces can be too large for low mass objects so using a smaller number might be desirable for some cases.
车轮反作用力用浮点数而不是布尔值启用。车轮力对低质量物体可能过大，因此某些情况下使用更小的数值更合适。

• You can try engine on/off feature in Modificaton.map in the sample project.
可以在示例工程的 Modificaton.map 中试用发动机开关功能。

• Vehicles failed to stay on rotating objects, its fixed now.
载具以前无法留在旋转物体上，现已修正。

• Trace Complex option(FCollisionQueryParams::bTraceComplex) added to tracing options. Enable to trace against complex(per poly) collision.
追踪选项增加了 Trace Complex（FCollisionQueryParams::bTraceComplex）。启用后会对复杂（逐多边形）碰撞进行追踪。

• In some cases throttling failed to generate wheel torque when the vehicle goes backwards while in 1st gear, this is fixed now.
某些情况下，载具在一挡向后行驶时踩油门无法产生车轮扭矩，现已修正。

• Skidmarks were generating warnings in 4.27, it's fixed now.
轮胎印在 4.27 中会产生警告，现已修正。

• Ray gizmos are not displayed if trace mode is not Line.
追踪模式不是 Line 时不显示射线 Gizmo。

• All UPROPERTY tags are converted to FPROPERTY as we dropped 4.24 support.

由于不再支持 4.24，全部 UPROPERTY 标记已转换为 FPROPERTY。

• All maps in sample project used to have 2x2km box brushes, we used smaller blocks(400x400m) to avoid floting point issues with convex casting.
示例工程中的地图以前都有 2×2 公里的盒体笔刷，我们改用更小的块（400×400 米），以避免凸体投射的浮点问题。

• Motion blur is disabled for all maps in sample project.
示例工程中的所有地图都关闭了运动模糊。

• DragRace.map in sample project is modified to have a finish line.
示例工程中的 DragRace.map 已修改，增加了终点线。

v1.4.2
v1.4.2

• Override input tick event
覆盖输入的 Tick 事件

• Articulated vehicle sample
铰接载具示例

• Fixed changing gear count at runtime bug
修正了运行时修改挡位数量的错误

• CustomCollision hide categories are removed
移除了 CustomCollision 的隐藏分类

Notes
说明

• In order to override standard input you had to set inputs from level BP. Overriding input from vehicle BP was not possible but you can override "OverrideInputTick" function now. Check Sedan_BP and Modification.map for an example.
以前要覆盖标准输入，必须从关卡蓝图设置输入。无法从载具蓝图覆盖输入，但现在可以覆盖“OverrideInputTick”函数。示例见 Sedan_BP 和 Modification.map。

• Sandbox map includes a new articulated vehicle. The skeletal mesh consist of vehicle body and the bed. There is also a tailgate attached which is a separate static mesh component. Both the bed and the tailgate are connected with a physics constraint. Hold middle mouse button down to raise the bed and release to lower it.
沙盒地图包含一辆新的铰接载具。骨骼网格由车身和货厢组成。还有一块作为独立静态网格组件连接的尾门。货厢和尾门都用物理约束连接。按住鼠标中键升起货厢，松开则降下。

• Changing gear count at runtime was problematic due to a simple bug, it is fixed now.
运行时修改挡位数量以前因为一个简单错误而出问题，现已修正。

• CustomCollision component had hide category tags with the intention of a clean interface but it seems that can be limiting so we removed them all.
CustomCollision 组件曾带有隐藏分类标签，本意是让界面更干净，但这可能造成限制，因此已全部移除。

v1.4.1
v1.4.1

• Auto clutch option for transmission
变速箱的自动离合器选项

• Fixed sequential gearbox bug
修正了序列式变速箱的错误

• Force symmetry option for tire models

轮胎模型的强制对称选项

Notes
说明

• There was a bug preventing sequential gear change, it is fixed now.
以前有一个错误阻止序列式换挡，现已修正。

• Auto clutch is on by default which enables you to change gears like an f1 gearbox. When auto clutch is off you can use clutch manually with sequential gearbox like a motorcycle.
自动离合器默认开启，因此可以像 F1 变速箱那样换挡。关闭自动离合器后，可以像摩托车那样配合序列式变速箱手动使用离合器。

• Tire model parameters can generate asymmetric curves and it can rarely cause problems. Force symmetry option forces the tire model to generate a symmetric curve by mirroring the y-axis of the curve. Keep it false(default) unless you encounter asymmetric steering behaviors.
轮胎模型参数可能生成不对称曲线，偶尔会造成问题。强制对称选项会镜像曲线的 Y 轴，迫使轮胎模型生成对称曲线。除非遇到不对称的转向表现，否则保持 false（默认）。

v1.4
v1.4

• Suspension constraint component
悬架约束组件

• Rewind and replay sample
倒带和回放示例

• Improved sticky tires
改进了低速黏附轮胎

• Suspension relaxation damper option
悬架回弹阻尼选项

• Clutch power scale option
离合器动力缩放选项

• Minimap and ranking UI samples
小地图和排名界面示例

• Default physical material option
默认物理材质选项

• Improvements and bug fixes
改进与错误修复

Notes
说明

• From now on it is strongly recommended to keep your root bone location to zero origin(0,0,0).
从现在起，强烈建议把根骨骼位置保持在零原点 (0,0,0)。

• Suspension constraint adds a joint to each wheel and updates their limits at each tick. This works much better then "Hard Contact" option so this is the recommended solution now.
悬架约束为每个车轮添加一个关节，并在每个 Tick 更新其限制。这比 “Hard Contact” 选项好得多，因此现在是推荐方案。

• You can use saveState/loadState methods to store and restore a vehicles state, FGearRewindReplay actor uses this methods.
可以用 saveState/loadState 方法保存和恢复载具状态，FGearRewindReplay Actor 使用这些方法。

• FGearRewindReplay may not be suitable for production since it is a framerate dependent solution.
FGearRewindReplay 可能不适合正式产品，因为它是依赖帧率的方案。

• Sticky tire state is represented as a float instead of bool now. This value is used to scale tire forces to get a more consistent behavior.
低速黏附轮胎状态现在用浮点数而不是布尔值表示。该值用来缩放轮胎力，以得到更一致的表现。

• Suspension damper param is divided into compression and relaxation damper params.
悬架阻尼参数被分成压缩阻尼和回弹阻尼两个参数。

• If "Suspension Relexation Downforce" is checked, relaxation damper can generate forces that can pull the vehicle towards the ground.
若勾选 “Suspension Relexation Downforce”，回弹阻尼可以产生把载具拉向地面的力。

• "Allow Wheel Rotation Against Engine" option had a bug that caused vehicles to move forward without throttling, it is fixed now.
“Allow Wheel Rotation Against Engine” 选项有一个错误，会使载具在没有油门时向前移动，现已修正。

• Clutch power scale can result jittery behavior if you use larger values.
离合器动力缩放如果使用较大的值，可能导致抖动。

• Feedback torque calculation is clamped to avoid weak responses caused by tire model problems.
反馈扭矩的计算被钳制，以避免轮胎模型问题造成的响应过弱。

• Rear gear max speed is also displayed in custom ui now.
自定义界面现在也会显示倒挡最高速度。

• All vehicle components are accessible from(private access) BP now.
现在可以从蓝图访问全部载具组件（私有访问）。

• We added more getter/setters for the extra components.
我们为扩展组件增加了更多 getter 和 setter。

• "gearChanged" event is fired from FGearTransmission now.
现在由 FGearTransmission 触发 “gearChanged” 事件。

• Telemetry is also displayed on clients now.
遥测现在也会显示在客户端上。

• FGearEffects will use the "Default Physical Material" if a surface has no physical material assigned.
若表面没有指定物理材质，FGearEffects 会使用 “Default Physical Material”。

• The last used vehicle in EnterExit demo will switch to parking mode(handbrake on) when unpossesed.
EnterExit 演示中最后使用的载具在取消 Possess 时会切换到驻车模式（手刹拉起）。

• OrbitCamera component active state is checked so that you can switch to alternate cameras now.
会检查 OrbitCamera 组件的激活状态，因此现在可以切换到其他摄像机。

• We added a settings category under ProjectSettings->Game->FgearCommon. Currently you can only override replication settings for all vehicles here.
我们在 Project Settings → Game → FgearCommon 下增加了一个设置分类。目前只能在这里覆盖所有载具的复制设置。

v1.3.1
v1.3.1

• Better hard contact implementation
更好的硬接触实现

• Ability to look back with OrbitCamera
可以用 OrbitCamera 向后看

• Fixed the low fps suspension bug
修正了低帧率下的悬架错误

Notes
说明

• Previous hard contact implementation generated larger forces when the suspension is fully compressed which could cause bouncy behaviour. Now the vehicle is translated to a non-penetrating position and the vertical velocity is eliminated.
以前的硬接触实现会在悬架完全压缩时产生更大的力，可能导致弹跳。现在会把载具平移到不穿透的位置，并消除竖直速度。

• Press the middle mouse button to look back in the sandbox map.
在沙盒地图中按鼠标中键向后看。

• Using the auto drive component with an empty brake curve could crash the engine, it is fixed now.
自动行驶组件使用空的制动曲线时可能使引擎崩溃，现已修正。

• The suspensions started to break when the frame rate falls below a certain value, it is fixed now.
帧率降到某个值以下时悬架会开始损坏，现已修正。

• "Detect Ground Velocity" option is added to the serializer.
序列化器增加了 “Detect Ground Velocity” 选项。

v1.3
v1.3

• Server authorative replication(no client prediction yet)
服务器权威复制（还没有客户端预测）

• Motocycle sample with BikeHelper assist(ArcadeAssists)
带 BikeHelper 辅助（ArcadeAssists）的摩托车示例

• New DragRace map for the sample project
示例工程的新 DragRace 地图

• Substep tick event available for blueprints
蓝图可以使用子步 Tick 事件

• Ability to drive on moving rigidbodies
可以在运动的刚体上行驶

• Improvements and bug fixes
改进与错误修复

Notes
说明

• "Allow Wheel Rotation Against Engine" option had a problem that prevented vehicles from stopping on flat surfaces, this is fixed now.
“Allow Wheel Rotation Against Engine” 选项有一个问题，会使载具无法在平地上停住，现已修正。

• Torque splitter assist could be problematic with low frame rates, this is fixed now.
扭矩分配辅助在低帧率下可能出问题，现已修正。

• Sphere casting could fail to generate a contact in some cases, this is fixed now.
球体投射在某些情况下无法产生接触，现已修正。

• There was a bug that effected 4wd vehicles that have different radius for each axle(like a tractor), this is fixed now.
有一个错误会影响各轴半径不同的四驱载具（例如拖拉机），现已修正。

• Instant suspension contacts could generate huge forces, resulting forces are clamped now.
瞬时悬架接触可能产生巨大的力，现在会对结果力进行钳制。

• Antirollbars in arcade assists were buggy, they are adapted to the current suspension implementation.
街机辅助中的防倾杆有错误，已按当前悬架实现做了适配。

• There exists some self steering issues, one of them is caused by the "(Mx)Overturn Moment Scale" parameter and it happens at high speeds like 300+ kmh. Another issue can occur because of the "Combined Lateral" options of a tire model. Setting some parameters(RVY1-6) to zero can solve the issue.
存在一些自行转向的问题。其中一个由 “(Mx)Overturn Moment Scale” 参数引起，发生在 300 km/h 以上的高速。另一个可能由轮胎模型的 “Combined Lateral” 选项引起。把某些参数（RVY1 到 RVY6）设为 0 可以解决。

• We used to enable/disable standard input component when the vehicle is possesed/unpossesed. This is removed now so the user has to do that manually. The sandbox maps level blueprint is adapted to this change.
我们以前会在载具被 Possess 或取消 Possess 时启用或禁用标准输入组件。现在去掉了这一行为，因此用户必须手动处理。沙盒地图的关卡蓝图已按此修改。

• In some rare cases the standard input component could cause a random crash. We suspected of a garbage collection issue and updated the UInputComponent as an UPROPERTY. The problems seems to be resolved now.
在一些少见情况下，标准输入组件可能随机崩溃。我们怀疑是垃圾回收问题，并把 UInputComponent 更新为 UPROPERTY。问题现在看来已经解决。

• Skidmark normals were based on vehicle up direction, replaced with surface normal now.
轮胎印法线以前基于载具的向上方向，现在改为表面法线。

• You can inject inputs to the StandardInput component now, for ex. in MobileInput.map inputs are not directly applied to the vehicle but injected to the StandardInput now. This way you can benefit from its filters and steering assist.
现在可以把输入注入 StandardInput 组件。例如在 MobileInput.map 中，输入不再直接作用到载具，而是注入到 StandardInput。这样就能使用它的过滤和转向辅助。

• DragRace map showcases the use of "SetStartGridMode" function in StandardInput component.
DragRace 地图展示了 StandardInput 组件中 “SetStartGridMode” 函数的用法。

• Each assist in ArcadeAssists component has enable/disable option now, they are all off by default.
ArcadeAssists 组件中的每项辅助现在都有启用/禁用选项，默认全部关闭。

• 4.22 support is dropped, epic do not allow more then 3 versions to be uploaded.
不再支持 4.22，Epic 不允许上传超过 3 个版本。

Replication Notes
复制说明

• Replication component implements input and state synchronization, client prediction is not working correctly yet(you will feel the lag) so it is delayed for the next release.
复制组件实现了输入同步和状态同步，客户端预测还不能正确工作（你会感觉到延迟），因此推迟到下一个版本。

• Bandwidth optimization was also not the focus of this release, the result of client prediction will show us if we can compress inputs and/or state data.
带宽优化也不是这个版本的重点。客户端预测的结果将说明我们能否压缩输入和/或状态数据。

• Projective velocity blending is preferred over interpolation for state synchronization, it shows the vehicles closer to their real locations but it can suffer from spikes in frame or ping rates.
状态同步优先使用投影速度混合，而不是插值。它让载具更接近真实位置，但会受帧率或延迟尖峰影响。

• Replication does not work properly with 32 bit windows builds, the reason is unknown but somehow

rotations(FQuat) are not replicated.
复制在 32 位 Windows 构建上不能正常工作，原因未知，但旋转（FQuat）不知为何没有被复制。

• When testing replication(listen server) with separate windows(different processes with multiple displays) the window that loses focus becomes erratic in framerate so it seems like it does not work smooth but this does not happen with 4.22 or with a dedicated server.
用分开的窗口（多个显示器上的不同进程）测试复制（监听服务器）时，失去焦点的窗口帧率会变得不稳定，因此看起来不平滑；但在 4.22 或专用服务器上不会这样。

• Stable framerate is important so the best is to have a fixed framerate or at least limited to some value like 60fps.
稳定的帧率很重要，因此最好使用固定帧率，或至少限制到 60 帧这类数值。

• When using a dedicated server make sure you limit the tick rates or you will experience huge delays, you can add the following lines to the DefaultEngine.ini file:
使用专用服务器时，请务必限制 Tick 频率，否则会有很大延迟。可以把下面几行加到 DefaultEngine.ini：

*[/Script/OnlineSubsystemUtils.IpNetDriver]*
[/Script/OnlineSubsystemUtils.IpNetDriver]：网络驱动配置段，需原样写入。

*+MaxNetTickRate=60*
+MaxNetTickRate=60：最大网络 Tick 频率为 60。

*+NetServerMaxTickRate=60*
+NetServerMaxTickRate=60：网络服务器最大 Tick 频率为 60。

*+LanServerMaxTickRate=60*
+LanServerMaxTickRate=60：局域网服务器最大 Tick 频率为 60。

*+NetClientTicksPerSecond=60*
+NetClientTicksPerSecond=60：网络客户端每秒 Tick 数为 60。

*+bClampListenServerTickRates=true*
+bClampListenServerTickRates=true：钳制监听服务器的 Tick 频率。

v1.2
v1.2

• Substepping support
支持物理子步

• Save/Load option(editor only)
保存/加载选项（仅编辑器）

• Sticky tire option for preventing low speed slips (experimental)
低速黏附轮胎选项，用于防止低速滑动（试验性）

• Custom editor UI elements (for ex. you can see top speeds for each gear in editor)
自定义编辑器界面元素（例如可以在编辑器中看到每个挡位的最高速度）

• Basic AI Driver component
基础 AI 驾驶员组件

• New AITest map for sample project
示例工程的新 AITest 地图

• Separate axis mode option for StandardInput
StandardInput 的分轴模式选项

• Option to prevent tire-ground penetration

防止轮胎穿透地面的选项

• Sphere casting option
球体投射选项

• An additional rc car setup for the sandbox map
沙盒地图增加了一套遥控车配置

• Minor improvements and bug fixes
少量改进与错误修复

Notes
说明

• Substepping is the recommended approach now.
现在推荐使用物理子步。

• Due to a problem with the unreal editor, it can be problematic to save/load from the blueprint editor window. When blueprint window is open and the viewport tab is not selected, extra axle components(axles other then front/rear) can not be retrieved so it is safer to save/load from regular details panel or make sure the viewport panel is active in blueprint window.
由于虚幻编辑器的一个问题，从蓝图编辑器窗口保存/加载可能有问题。蓝图窗口打开且没有选中视口页签时，取不到额外的车轴组件（前轴/后轴以外的车轴），因此更安全的做法是从普通细节面板保存/加载，或确保蓝图窗口中的视口面板处于激活状态。

• If you use separate axis for throttle/brake then you can throttle & brake at the same time.
如果油门和制动使用分开的轴，就可以同时踩油门和制动。

• You can call ArcadeAssists::setCruiseSpeed method to set the vehicles instant speed. This could be usefull for rolling starts.
可以调用 ArcadeAssists::setCruiseSpeed 设置载具的瞬时速度。这可用于滚动发车。

• Rc cars animation blueprint is copied from unreals own vehicle sample.
遥控车的动画蓝图复制自虚幻自带的载具示例。

• An additional parameter called RimOffset is added, this will offset the physical wheels laterally in case the visuals do not match(Rc car is an example).
增加了名为 RimOffset 的参数，它会横向偏移物理车轮，用于视觉与物理不重合的情况（遥控车就是一个例子）。

• If you prevent tire-ground penetration, tires can penetrate the fenders.
如果防止轮胎穿透地面，轮胎就可能穿透翼子板。

• You can change every value of WheelOptions for each wheel now. Changing these values for left and right wheel can be problematic in some cases where there is an assumption that left/right wheels have the same value. For ex. if left and right wheels have different radius and the difference is more then acceptable ranges then the transmission will change gears earlier then expected because gear changes are based on the slowest wheel.
现在可以修改每个车轮的 WheelOptions 的每一项数值。在某些假定左右车轮数值相同的情况下，为左右轮设置不同数值可能出问题。例如左右轮半径不同，且差值超出可接受范围时，变速箱会比预期更早换挡，因为换挡以最慢的车轮为准。

v1.1
v1.1

• Network replication
网络复制

• Suspension improvements including hard contact option
悬架改进，包括硬接触选项

• Auto reverse option for transmission
变速箱的自动倒挡选项

• Sample backfire implementation for FGearEffects component
FGearEffects 组件的示例回火实现

• Runtime modification map for example project
示例工程的运行时修改地图

• Enter/exit vehicle map for example project
示例工程的上下车地图

• Stress test map for example project
示例工程的压力测试地图

• Multiplayer map for example project
示例工程的多人地图

• Two additional sample setups for the sandbox map
沙盒地图增加了两套示例配置

• Minor improvements and bug fixes
少量改进与错误修复

Notes
说明

• For network replication we tried to use regular pawn replication and tried to make it work along with physics. We only send/receive user input in addition to pawn replication so the effects are triggered like a local game which makes things easier. Depending on your feedback(if it is problematic) we may rewrite our current approach in the future releases.
网络复制方面，我们尝试使用常规的 Pawn 复制，并让它与物理一起工作。除了 Pawn 复制，我们只发送和接收用户输入，因此效果会像本地游戏一样触发，这让事情更简单。根据反馈（如果它有问题），我们可能在以后的版本中重写当前做法。

• About the multiplayer map you can set player count and test directly or tick dedicated server option and pick the server and join by entering ip address.
关于多人地图，可以直接设置玩家数量进行测试，也可以勾选专用服务器选项，选择服务器，并输入 IP 地址加入。

• You can set the vehicle mass, center of mass offset and inertia scale at runtime with this version.
这个版本可以在运行时设置载具质量、质心偏移和惯量缩放。

• You can assign a new torque curve to the engine at runtime with this version.
这个版本可以在运行时为发动机指定一条新的扭矩曲线。

• You can get max engine power(kw) and max power rpm with this version.
这个版本可以取得发动机最大功率（千瓦）和最大功率转速。

• You can get max speed for each gear with this version.
这个版本可以取得每个挡位的最高速度。